Abstract

Keywords:

1. Introduction

2. Related Work

2.1. Pedestrians in the loop and VR experiments
2.2. Low-vision simulation and rehabilitation in VR
2.3. Scenario specification languages and ontologies
2.4. LLM-based scenario generation
2.5. Pedestrian-focused scenario generation for autonomous-vehicle testing
2.6. Urban and 3D scene generation
2.7. Positioning and research gap

3. Problem Statement and Research Questions

4. Pedestrian Navigation Task Taxonomy and Representation

4.1. Scope and inclusion criteria
4.2. Task categories
4.3. Task ontology: targets and failure conditions
4.4. Encoding in OpenSCENARIO and OpenDRIVE

5. System Overview

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5.2. LLM agent and grounded tools
5.3. Scenario DSL and deterministic storyboard assembly
5.4. Sandboxed execution
5.5. Map-aware validation and repair loop
5.6. Interactive player (VR and first person)

6. Evaluation

6.1. Experimental setup
6.2. Benchmark: validity and prompt adherence (RQ2, RQ3)

Table 1: Performance of the L2 pipeline per scenario type on the Sophia Antipolis map. Result: count (percentage; 95% Wilson interval) over all runs of that type.

Scenario type	Task description	Quantitative metric	Result
`ptp_env` (9 runs)	Reach a landmark (bus stop, bench, waste basket) over walkable ground	Accepted (validator passes the final scenario)	9/9 (100%; 70–100)
		First try (accepted with no repair loop)	3/9 (33%; 12–65)
		Adherent (constraint axes and operators match the request)	9/9 (100%; 70–100)
		Solvable (the nominal walker satisfies all tasks)	9/9 (100%; 70–100)
		Script submissions per run (mean)	1.33
		Model steps per run (mean)	3.7
		Wall-clock per run, s (mean)	44
`ptp_road` (9 runs)	Reach a spot on the carriageway	Accepted (validator passes the final scenario)	6/9 (67%; 35–88)
		First try (accepted with no repair loop)	4/9 (44%; 19–73)
		Adherent (constraint axes and operators match the request)	5/9 (56%; 27–81)
		Solvable (the nominal walker satisfies all tasks)	5/9 (56%; 27–81)
		Script submissions per run (mean)	0.89

Scenario type	Task description	Quantitative metric	Result
		Model steps per run (mean)	6.4
		Wall-clock per run, s (mean)	52
`cpath` (9 runs)	Reach a goal while staying inside a corridor (path or sidewalk)	Accepted (validator passes the final scenario)	1/9 (11%; 2–44)
		First try (accepted with no repair loop)	0/9 (0%; 0–30)
		Adherent (constraint axes and operators match the request)	1/9 (11%; 2–44)
		Solvable (the nominal walker satisfies all tasks)	1/9 (11%; 2–44)
		Script submissions per run (mean)	4.89
		Model steps per run (mean)	6.1
		Wall-clock per run, s (mean)	121
`env_con` (9 runs)	Reach a goal without touching static obstacles (parked cars)	Accepted (validator passes the final scenario)	4/9 (44%; 19–73)
		First try (accepted with no repair loop)	2/9 (22%; 6–55)
		Adherent (constraint axes and operators match the request)	4/9 (44%; 19–73)
		Solvable (the nominal walker satisfies all tasks)	4/9 (44%; 19–73)

Scenario type	Task description	Quantitative metric	Result
		Script submissions per run (mean)	3.11
		Model steps per run (mean)	6.8
		Wall-clock per run, s (mean)	185
`dyn_nav` (9 runs)	Cross while keeping clear of a moving vehicle	Accepted (validator passes the final scenario)	5/9 (56%; 27–81)
		First try (accepted with no repair loop)	2/9 (22%; 6–55)
		Adherent (constraint axes and operators match the request)	5/9 (56%; 27–81)
		Solvable (the nominal walker satisfies all tasks)	4/9 (44%; 19–73)
		Script submissions per run (mean)	4.33
		Model steps per run (mean)	5.6
		Wall-clock per run, s (mean)	185
`dep_comp` (9 runs)	Several goals that must be reached in a given order	Accepted (validator passes the final scenario)	5/9 (56%; 27–81)
		First try (accepted with no repair loop)	0/9 (0%; 0–30)
		Adherent (constraint axes and operators match the request)	5/9 (56%; 27–81)
		Solvable (the nominal walker satisfies all tasks)	5/9 (56%; 27–81)

Scenario type	Task description	Quantitative metric	Result
		Script submissions per run (mean)	4.44
		Model steps per run (mean)	8.2
		Wall-clock per run, s (mean)	192
`indep_comp` (9 runs)	Several goals that may be reached in any order	Accepted (validator passes the final scenario)	7/9 (78%; 45–94)
		First try (accepted with no repair loop)	0/9 (0%; 0–30)
		Adherent (constraint axes and operators match the request)	0/9 (0%; 0–30)
		Solvable (the nominal walker satisfies all tasks)	7/9 (78%; 45–94)
		Script submissions per run (mean)	2.22
		Model steps per run (mean)	10.2
		Wall-clock per run, s (mean)	139
`mixed` (12 runs)	Combined constraints: signal compliance, a moving car, looking at it	Accepted (validator passes the final scenario)	2/12 (17%; 5–45)
		First try (accepted with no repair loop)	0/12 (0%; 0–24)
		Adherent (constraint axes and operators match the request)	2/12 (17%; 5–45)

Scenario type	Task description	Quantitative metric	Result
		Solvable (the nominal walker satisfies all tasks)	0/12 (0%; 0–24)
		Script submissions per run (mean)	2.58
		Model steps per run (mean)	6.9
		Wall-clock per run, s (mean)	224

6.3. Vision ablation

6.4. Coverage analysis of the taxonomy (RQ1)

6.5. Qualitative feedback round (RQ4)

7. Discussion

7.1. Limitations

8. Conclusion

CRediT authorship contribution statement

Declaration of competing interest

Funding

Data availability

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Appendix A. Evaluation prompts

Table A.2: Evaluation prompts (Sophia Antipolis map, all requested object types exist on it).

Id	Type	Prompt
sa_ptp_env_1	ptp_env	Walk along the sidewalk to the nearest bus stop.
sa_ptp_env_2	ptp_env	The participant should find the bench nearby and walk up to it.
sa_ptp_env_3	ptp_env	Go to the waste basket and stop next to it.
sa_ptp_road_1	ptp_road	Walk out onto the road and stand on the marked spot just past a crossing.
sa_ptp_road_2	ptp_road	The participant must stand in the middle of the carriageway for a moment.
sa_ptp_road_3	ptp_road	Have the user walk onto the road surface at the far end of the street.
sa_cpath_1	cpath	Cross the road but stay inside the marked corridor along the crosswalk.
sa_cpath_2	cpath	Follow the marked path along the street without leaving it.
sa_cpath_3	cpath	Walk to the crossing using only the sidewalk; stepping off it means failure.
sa_env_con_1	env_con	Cross the road at the zebra crossing without touching the parked cars.
sa_env_con_2	env_con	Walk between the parked cars along the kerb and reach the other end without touching any of them.
sa_env_con_3	env_con	Avoid the parked vehicles along the street while you walk to the bus stop.
sa_dyn_nav_1	dyn_nav	Cross at the zebra crossing while a car approaches; avoid being hit.

Table A.2: Evaluation prompts (Sophia Antipolis map, all requested object types exist on it).

Id	Type	Prompt
sa_dyn_nav_2	dyn_nav	A bus drives along the street. Walk across the road without being hit by it.
sa_dyn_nav_3	dyn_nav	Cross the road with oncoming traffic and keep clear of the speeding car.
sa_dep_comp_1	dep_comp	First walk to the crossing, then cross the road, then continue to the spot on the other side.
sa_dep_comp_2	dep_comp	Do three things in order: go to the bus stop, then to the bench, then to the waste basket.
sa_dep_comp_3	dep_comp	Walk to the kerb, afterwards cross the road, and finally walk to the waste basket.
sa_indep_comp_1	indep_comp	Visit the waste basket and the bench, in any order.
sa_indep_comp_2	indep_comp	Find both the bench and the bus stop; the order does not matter.
sa_indep_comp_3	indep_comp	Reach a bench and a tree, either one first.
sa_mixed_1	mixed	Wait for the pedestrian light to turn green, then cross the road.
sa_mixed_2	mixed	Cross at the signalised crossing on green while a car comes down the street, and stay clear of the car.
sa_mixed_3	mixed	Look at the approaching car, then cross the road once it has passed.
sa_mixed_4	mixed	At the crossing wait until the light is green, look at the traffic light, and then cross.

References

[1] H.-Y. Wu, F. Robert, T. Fafet, B. Graulier, B. Passin-Cauneau, L. Sassatelli, M. Winckler, Designing guided user tasks in VR embodied experiences, Proceedings of the ACM on Human-Computer Interaction 6 (EICS) (2022) 1–24. doi:10.1145/3532208.